Hardware list

Everything the recorder was built and tested with.

Parts

Part	Qty	Specification used
Raspberry Pi 5	1	8 GB RAM; Raspberry Pi OS 64-bit (Debian 13), Python 3.13
Storage	1	NVMe SSD (any size); recordings go to tmpfs first, then here
USB camera	4	UVC, 2 MP (1920x1080), MJPEG 30 fps, USB 2.0
I2S microphone	2	InvenSense ICS-43434 MEMS, 24-bit I2S
OLED display	1	SSD1306, 128x64, I2C, address 0x3C
Push button	2	momentary, normally open
Rig frame	1	rigid plate holding the four cameras in a 2x2 grid
Wires	-	female-female jumper wires for the 40-pin header

Camera

Camera property	Value	Why it matters
Resolution / format	1920x1080 MJPEG, 30 fps	MJPEG is stored as-is, no re-encoding
Interface	USB 2.0, UVC (USB ID 32e6:9221)	bandwidth limits 1080p to two cameras per bus
Lens	wide angle, about 88 deg horizontal	focal length about 980-1000 px at 1920 wide
Focus	adjustable; autofocus disabled	do not turn the lens after calibrating
Exposure / gain	manual; gain 1.0-1.484 (up to 3.7x)	fixed values keep all cameras identical
Byte rate	about 35 MB/s for four cameras	varies 3-7x with scene brightness and noise

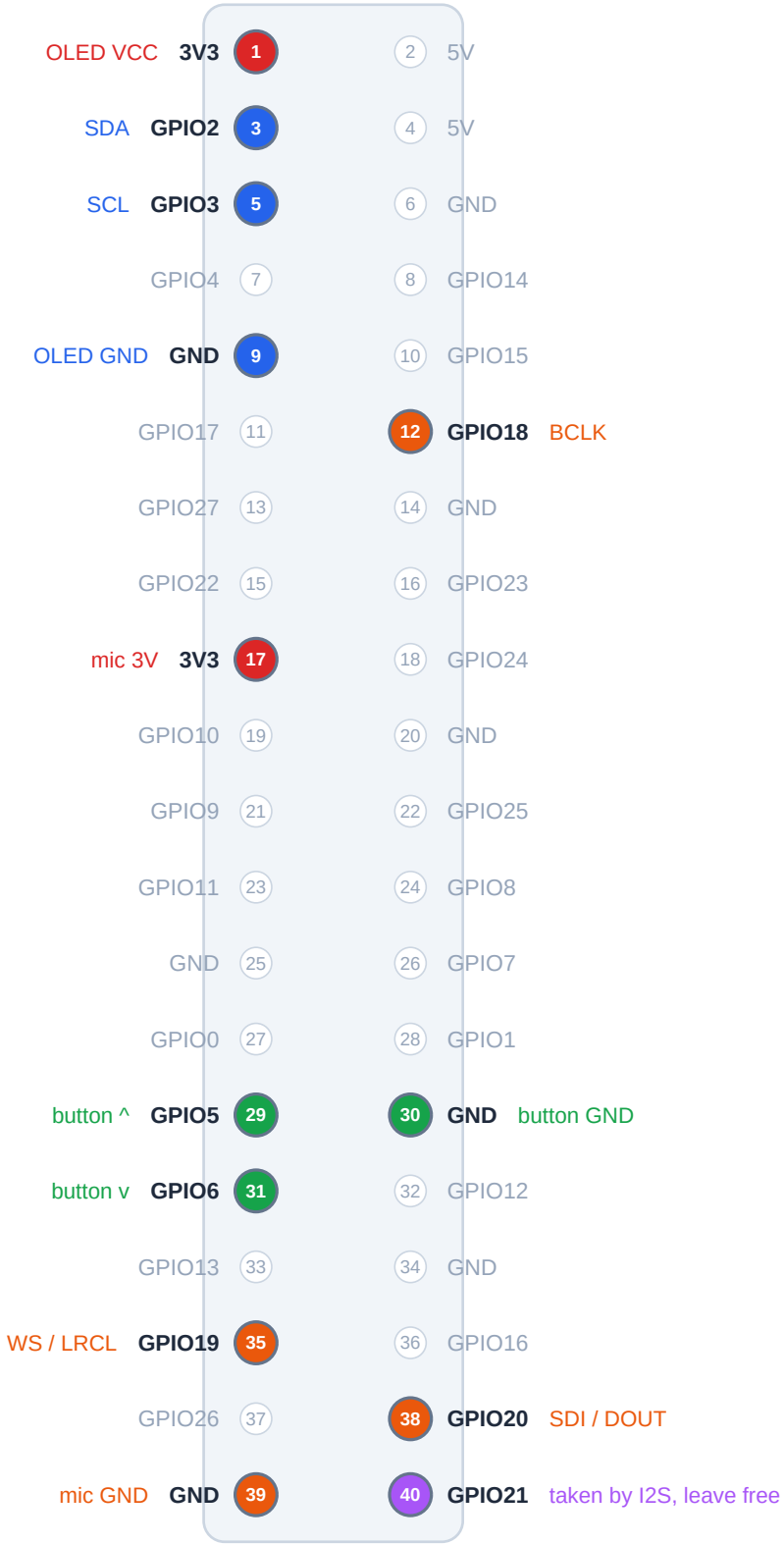
Microphones

Microphone property	Value
Type	ICS-43434, I2S digital MEMS, 24 valid bits in 32-bit frames
Sample rate	48 kHz (supported range 23.2-51.6 kHz; 44.1 and 96 kHz do not work)
Stereo	two microphones on one I2S bus; the SEL pin selects left or right
Spacing	195 mm apart (set mic_spacing_mm in config.json)

USB bandwidth: the Pi 5 has two USB controllers. Connect two cameras to each (the recorder warns when more share one bus). A second camera on the same bus already loses a few percent of frames under high gain, because noisy images make larger JPEGs. More light is the fix; raising gain makes it worse.

Pi 5 40-pin header

Pin 1 is on the side nearest the board edge. Only coloured pins are wired.



Legend

- OLED (I2C1), 4 wires
- buttons, 3 wires
- microphones (I2S)
- 3.3 V supply
- reserved, not wired
- free

Wiring, three groups

Each group can be wired and tested on its own.

1. OLED SSD1306 128x64 (I2C1)

OLED pin	Connect to	Header pin
VCC	3.3 V	1
GND	GND	9
SDA	GPIO 2	3
SCL	GPIO 3	5

OLED modules usually include pull-up resistors; none need to be added.

2. Two buttons

Button	One leg	Other leg
^ (upper)	GPIO 5 = pin 29	GND = pin 30
v (lower)	GPIO 6 = pin 31	GND = pin 30

Pins 29 / 30 / 31 are adjacent, so one 3-pin connector is enough. Both buttons share the GND on pin 30.

No external pull-ups: GPIO 0-8 are pulled up by default at boot, so the inputs never float before the program starts. This is why GPIO 5 and 6 were chosen.

3. Two ICS-43434 microphones (I2S, shared bus)

ICS-43434 pin	Left mic	Right mic	Header pin
3V	3.3 V	3.3 V	17
GND	GND	GND	39
SEL	GND	3.3 V	- selects left / right
BCLK	GPIO 18	GPIO 18	12
LRCL / WS	GPIO 19	GPIO 19	35
DOUT	GPIO 20	GPIO 20	38

BCLK, LRCL and DOUT are wired in parallel to both microphones; only SEL differs. Both drive the same DOUT line, each in its own half of the frame selected by SEL. That is how the two form a stereo pair.

GPIO 21 (pin 40) is not wired but cannot be used for anything else: the Pi 5 RP1 claims rp1_i2s0_18_21 as one group, so enabling I2S reserves GPIO 18-21.

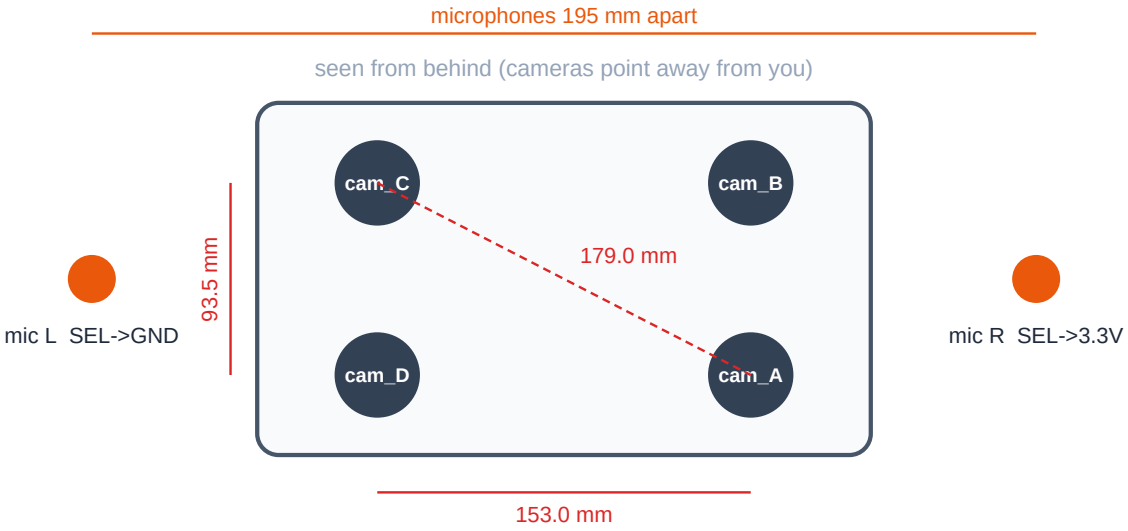
Boot configuration /boot/firmware/config.txt

```
dtparam=i2c_arm=on
dtparam=i2s=on
dtoverlay=googlevoicehat-soundcard
```

The overlay name looks unrelated, but it is the standard I2S stereo input setting on the Pi. Reboot after the change; `arecord -l` should then list the device.

Rig layout

Rear view: standing behind the rig, looking where the cameras look.



Measured camera spacing

Camera pair	Distance between centers (mm)
cam_C - cam_B, cam_D - cam_A (horizontal)	153.0
cam_C - cam_D, cam_B - cam_A (vertical)	93.5
cam_A - cam_C, cam_B - cam_D (diagonal)	179.0

Measure between lens centers, not housing edges. These six distances are passed to calibration (--edge) to give the reconstruction its metric scale; measure again whenever the rig is rebuilt.

Placement

Placement	Reason
Cameras in a rigid 2x2 grid, axes nearly parallel	the rig is the window frame; measured axes differ by 1.1-1.6 deg
Camera names follow USB ports	keep each camera on its port; moving one requires a new calibration
Microphones at the outer left and right edges	wider spacing gives a clearer stereo image
OLED and buttons on the back, via jumper wires	a HAT on the header would collide with the camera mount
Left / right as the cameras see	matches the calibration: cam_A is at +x (right), cam_C at -x, -y

Checks after wiring

What	Command / action	Expected
OLED detected	i2cdetect -y 1	3c (or 3d) appears
Buttons	gpioget gpiochip0 5 6	1 released, 0 pressed
Microphone device	arecord -l	the I2S card is listed
Sound captured	rig.py menu: mic test	both bars move
Left / right	mic test, tap the left microphone	only the L bar jumps

All checks must pass. If one fails, check that group's wires only; the three groups are independent.